

SQL Aggregate Functions



1. CREATE TABLE

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    department VARCHAR(50),  
    salary INT,  
    age INT,  
    experience INT,  
    city VARCHAR(50)  
);
```



2. INSERT INTO (Sample Data)

```
INSERT INTO employees VALUES  
(1, 'Amit', 'IT', 60000, 28, 4, 'Delhi'),  
(2, 'Ravi', 'HR', 45000, 32, 6, 'Mumbai'),  
(3, 'Neha', 'Finance', 55000, 30, 5, 'Delhi'),  
(4, 'Karan', 'IT', 70000, 35, 8, 'Bangalore'),  
(5, 'Pooja', 'HR', 40000, 26, 2, 'Pune'),  
(6, 'Arjun', 'Finance', 65000, 34, 7, 'Mumbai'),  
(7, 'Suman', 'IT', 50000, 29, 4, 'Delhi'),  
(8, 'Rohit', 'Sales', 48000, 31, 5, 'Chennai'),  
(9, 'Anita', 'Sales', 52000, 27, 3, 'Bangalore'),  
(10, 'Vikas', 'IT', 80000, 38, 10, 'Pune'),  
(11, 'Meena', 'Finance', 60000, 33, 6, 'Chennai'),  
(12, 'Deepak', 'HR', 42000, 29, 3, 'Delhi');
```



3. 50 Aggregate Function Queries (With Questions)



COUNT ()

1. Count total employees

```
SELECT COUNT(*) FROM employees;
```

2. Count employees in each department

```
SELECT department, COUNT(*) FROM employees GROUP BY department;
```

3. Count employees in each city

```
SELECT city, COUNT(*) FROM employees GROUP BY city;
```

4. Count employees with salary above 50,000

```
SELECT COUNT(*) FROM employees WHERE salary > 50000;
```

5. Count employees aged above 30

```
SELECT COUNT(*) FROM employees WHERE age > 30;
```

+ SUM()

6. Find total salary of all employees

```
SELECT SUM(salary) FROM employees;
```

7. Find total salary by department

```
SELECT department, SUM(salary) FROM employees GROUP BY department;
```

8. Find total salary by city

```
SELECT city, SUM(salary) FROM employees GROUP BY city;
```

9. Find total salary of IT department

```
SELECT SUM(salary) FROM employees WHERE department = 'IT';
```

10. Find total experience of all employees

```
SELECT SUM(experience) FROM employees;
```



AVG()

11. Find average salary of all employees

```
SELECT AVG(salary) FROM employees;
```

12. Find average salary department-wise

```
SELECT department, AVG(salary) FROM employees GROUP BY department;
```

13. Find average age of employees

```
SELECT AVG(age) FROM employees;
```

14. Find average experience by city

```
SELECT city, AVG(experience) FROM employees GROUP BY city;
```

15. Find average salary of Finance department

```
SELECT AVG(salary) FROM employees WHERE department = 'Finance';
```



MIN()

16. Find minimum salary

```
SELECT MIN(salary) FROM employees;
```

17. Find minimum salary in each department

```
SELECT department, MIN(salary) FROM employees GROUP BY department;
```

18. Find minimum age of employees

```
SELECT MIN(age) FROM employees;
```

19. Find minimum experience by department

```
SELECT department, MIN(experience) FROM employees GROUP BY  
department;
```

20. Find minimum salary in IT department

```
SELECT MIN(salary) FROM employees WHERE department = 'IT';
```

 MAX()

21. Find maximum salary

```
SELECT MAX(salary) FROM employees;
```

22. Find maximum salary department-wise

```
SELECT department, MAX(salary) FROM employees GROUP BY department;
```

23. Find maximum age

```
SELECT MAX(age) FROM employees;
```

24. Find maximum experience by city

```
SELECT city, MAX(experience) FROM employees GROUP BY city;
```

25. Find maximum salary in Sales department

```
SELECT MAX(salary) FROM employees WHERE department = 'Sales';
```

 GROUP BY + HAVING

26. Departments having more than 2 employees

```
SELECT department, COUNT(*)  
FROM employees  
GROUP BY department  
HAVING COUNT(*) > 2;
```

27. Cities having total salary above 1,00,000

```
SELECT city, SUM(salary)  
FROM employees  
GROUP BY city  
HAVING SUM(salary) > 100000;
```

28. Departments with average salary above 55,000

```
SELECT department, AVG(salary)  
FROM employees  
GROUP BY department  
HAVING AVG(salary) > 55000;
```

29. Cities having more than 2 employees

```
SELECT city, COUNT(*)  
FROM employees  
GROUP BY city  
HAVING COUNT(*) > 2;
```

30. Departments with minimum salary above 45,000

```
SELECT department, MIN(salary)  
FROM employees  
GROUP BY department  
HAVING MIN(salary) > 45000;
```



MIXED AGGREGATES

31. Total salary and average salary

```
SELECT SUM(salary), AVG(salary) FROM employees;
```

32. Count and average age

```
SELECT COUNT(*), AVG(age) FROM employees;
```

33. Department-wise max and min salary

```
SELECT department, MAX(salary), MIN(salary)
FROM employees
GROUP BY department;
```

34. City-wise total and average salary

```
SELECT city, SUM(salary), AVG(salary)
FROM employees
GROUP BY city;
```

35. Department-wise count and total experience

```
SELECT department, COUNT(*), SUM(experience)
FROM employees
GROUP BY department;
```



ADVANCED PRACTICE

36. Highest salary in IT department

```
SELECT MAX(salary) FROM employees WHERE department = 'IT';
```

37. Lowest salary in HR department

```
SELECT MIN(salary) FROM employees WHERE department = 'HR';
```

38. Average experience of employees older than 30

```
SELECT AVG(experience) FROM employees WHERE age > 30;
```

39. Total salary of employees younger than 30

```
SELECT SUM(salary) FROM employees WHERE age < 30;
```

40. Count employees with experience more than 5 years

```
SELECT COUNT(*) FROM employees WHERE experience > 5;
```

 FINAL SET

41. Department-wise employee count sorted

```
SELECT department, COUNT(*)  
FROM employees  
GROUP BY department;
```

42. City-wise max salary

```
SELECT city, MAX(salary)  
FROM employees  
GROUP BY city;
```

43. Average salary of employees in Delhi

```
SELECT AVG(salary) FROM employees WHERE city = 'Delhi';
```

44. Total experience by department

```
SELECT department, SUM(experience)  
FROM employees  
GROUP BY department;
```

45. Count employees in Pune

```
SELECT COUNT(*) FROM employees WHERE city = 'Pune';
```

46. Department with highest total salary

```
SELECT department, SUM(salary)  
FROM employees  
GROUP BY department;
```

47. City with lowest average salary

```
SELECT city, AVG(salary)
FROM employees
GROUP BY city;
```

48. Departments with more than 1 employee

```
SELECT department, COUNT(*)
FROM employees
GROUP BY department
HAVING COUNT(*) > 1;
```

49. Employees count and max age

```
SELECT COUNT(*), MAX(age) FROM employees;
```

50. Overall salary statistics

```
SELECT MIN(salary), MAX(salary), AVG(salary), SUM(salary)
FROM employees;
```